NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

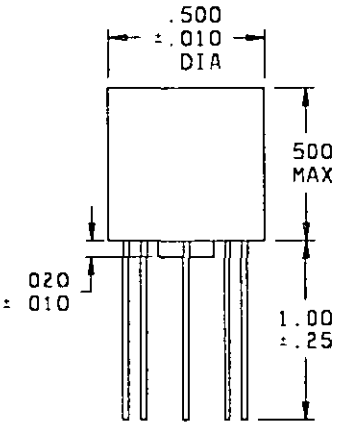
MIL-PRF-27/358B
14 June 1988
SUPERSEDING
MIL-T-27/358A
12 May 1987

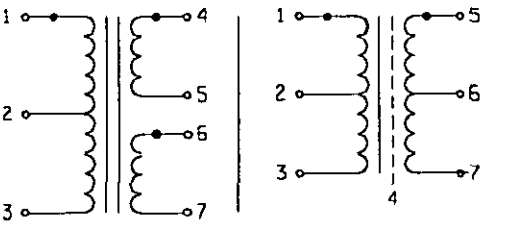
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY, SUB-MINIATURE, TF5R21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

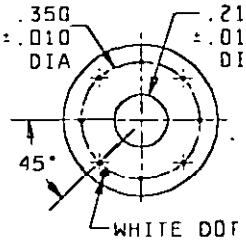
The complete requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-27.





WORKING VOLTAGE (PEAK): 100 V
ALTITUDE: 150,000 FT MAX

CIRCUIT DIAGRAM



WHITE DOT AT TERMINAL NO 1
TERMINALS CLOCKWISE FROM
PIN NO 1.

Inches	mm
.010	0.25
.020	0.51
.218	5.54
.350	6.4
.500	8.89
1.00	25.4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.
4. Primary and secondary electrical values, specified in table I, shall be marked as applicable.
5. Dots on circuit diagram indicate polarity.

FIGURE 1. Dimensions and configuration.

(B) denotes changes

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REQUIREMENTS:

Electrical ratings: See table I.

Working voltage (peak): 100 volts.

Frequency range: 150 Hz to 75 kHz.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Terminals: Solid wire.

Composition: Copper clad steel.

Diameter: 0.016 inch ±0.001 (0.41 ±0.03 mm).

Length: 1.00 inch ±0.25 (25.4 ±6.35 mm).

Final finish: Tinned, 100 percent, electroplated in accordance with MIL-T-10727.

Weight: 7.5 grams.

Altitude: 150,000 feet, maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Thermal shock: 25 cycles, MIL-STD-202, method 107, test condition A-1.

Vibration (high frequency): MIL-STD-202, method 204, test condition B.

Dielectric withstanding voltage:

At sea level: 200 V rms.

At barometric pressure: 200 V rms.

Electrical characteristics:

No load (center tap unbalance): ±1 percent at 1 volt, 5 kHz across the primary.

Harmonic distortion: Total harmonic content of the output shall be a maximum of 5 percent at the specified power level (see table I) at 1 kHz, 400 Hz, and 150 Hz.

Insertion loss: At the specified power level (see table I), the insertion loss shall be a maximum of 3.0 dB at 1 kHz.

Frequency response: ±3 dB at the rated source and load impedances (see table I) with 1.0 mW level output and a reference frequency of 1 kHz over a frequency range of 150 Hz to 75 kHz.

Polarity: Additive when appropriate terminals are connected. For configuration A, connect terminals 3 and 4, 5 and 6. For configuration B, connect terminals 3 and 5. (See figure 1, configurations A and B).

Ⓑ Self-resonance: The self-resonant frequency shall be greater than 150 kHz when measured with the secondaries carrying the specified load and the secondary voltage observed.

Marking location: See figure 1.

Part number: M27/358- (dash number from table I).

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TABLE I. Electrical ratings. 1/

Dash no.	Primary impedance (ohms)	Secondary impedance (ohms)		Power watts			Primary unbalanced dc current (mA)	Primary dc resistance (ohms)	Primary balanced current (mA)	Primary dc resistance (ohms)	Secondary dc resistance (ohms)		Turns ratio primary/secondary		Circuit diagram configuration
		Series	Parallel	1 kHz and higher	400 Hz and higher	150 Hz and higher					Series connection	Parallel connection	Secondary series connection	Secondary parallel connection	
[illegible]	20 ct	8 ct	2	1.0	.75	.11	22 0	1 75	22 0	1 75	1.0	25	1 58 1	3.16.1	A
[illegible]	20 ct	20 ct	5	1 0	75	11	22 0	1 75	22.0	1 75	2 5	63	1 1	2 1	"
003	32 ct	32 ct	8	1 0	75	.11	18.0	2.7	18 0	2.7	3.7	.93	1 1	2·1	"
004	40 ct	40 ct	10	1.0	.75	11	16.0	3	16 0	3	5.2	1 3	1 1	2·1	"
005	48 ct	16 ct	4	1 0	75	11	13 0	4 5	13 0	4 5	2.4	6	1.73:1	3 46 1	"
006	50 ct	50 ct	12.5	1.0	.75	11	13 0	4 5	13.0	4.5	6.2	1.6	1 1	2.1	"
007	80 ct	32 ct	8	1 0	.75	.11	11.0	6 6	11.0	6.6	3.7	93	1.58 1	3.16 1	"
008	100 ct	40 ct	10	1.0	75	.11	10.0	9 7	10.0	9.7	5.6	1 4	1 58·1	3 16·1	"
009	100 ct	100 ct	25	1.0	.75	.11	10.0	9.7	10.0	9 7	11	2.8	1 1	2 1	"
010	120 ct	12 8 ct	3.2	1.0	.75	.11	9.0	10 8	9 0	10 8	1 6	4	3 06 1	6.12·1	"
011	120 ct	80 ct	15	1 0	75	11	9.0	10 8	9 0	10.8	8.5	2.2	1.41:1	2.83:1	"
012	120 ct	120 ct	30	1.0	.75	11	9.0	10 8	9 0	10.8	15	3.8	1:1	2:1	"
013	150 ct	12 ct	3	1.0	.75	.11	8.0	13.5	8.0	13.5	1.75	.44	3 54·1	7.08·1	"
014	150 ct	50 ct	12.5	1 0	75	11	8 0	12	8 0	12	5 8	1 5	1 73·1	3 46 1	"
015	150 ct	100 ct	25	1 0	75	11	8 0	12	8 0	12	13	3 3	1.22:1	2.44:1	"
016	150 ct	150 ct	37.5	1.0	.75	.11	8 0	12	8 0	12	20	5	1·1	2:1	"
017	150 ct	600 ct	150	1.0	.75	.11	8.0	12	8.0	12	82	20.5	1·2	1·1	"
018	200 ct	200 ct	50	1.0	.75	11	7.0	17	7.0	17	24	6 0	1 1	2.1	"
019	300 ct	12 ct	3	1.0	75	.11	5 0	30	5 0	30	1.75	44	9·1	10:1	"
020	300 ct	50 ct	12 5	1 0	75	11	5 0	27	5 0	27	6.2	1.6	2.45:1	4.9:1	"
021	300 ct	150 ct	37 5	1.0	75	.11	5.0	26	5.0	26	21	5.3	1 41:1	2 83 1	"
022	300 ct	300 ct	75	1.0	.75	11	5 0	27	5 0	27	38	9 5	1·1	2:1	"
023	300 ct	800 ct	150	1 0	.75	11	5.0	27	5.0	27	66	17	1:1.41	1.41:1	"
024	320 ct	12 8 ct	3 2	1 0	.75	11	5.0	30	5.0	30	1.75	.44	8:1	10.1	"
025	400 ct	40 ct	10	1.0	.75	.11	5.0	39	5.0	39	4.3	1.1	3.16:1	6.32:1	"
026	400 ct	120 ct	30	1.0	.75	11	5.0	30	5 0	30	14	3.5	1·83.1	3 65·1	"
027	400 ct	200 ct	50	1 0	75	.11	5 0	39	5 0	39	24	6	1 41:1	2.83:1	"
028	400 ct	400 ct	100	1.0	.75	11	5 0	39	5.0	39	42	10.5	1.1	2:1	"
029	400 ct	1000 ct	250	1 0	75	.11	5.0	39	5.0	39	109	28	1 1.58	1.26:1	"
030	400 ct	2000 ct	500	1.0	75	11	5.0	39	5.0	39	198	49	1 2.24	1 1.12	"
031	400 ct	4000 ct	1000	1.0	.75	11	5 0	39	5 0	39	436	109	1 3.16	1:1.58	"
032	500 ct	12 8 ct	3 2	1.0	75	11	4 5	43	4 5	43	1.6	4	6.25.1	12.5:1	"
033	500 ct	16 ct	4	1.0	75	.11	4.5	43	4.5	43	2 3	.56	5 6:1	11.2:1	"
034	500 ct	50 ct	12 5	1.0	.75	.11	4 5	43	4.5	43	5.8	1.5	3.16:1	6.32.1	"
035	500 ct	150 ct	37.5	1.0	75	.11	4 5	43	4.5	43	19.8	5	1 83:1	3.65:1	"
036	500 ct	500 ct	125	1.0	.75	.11	4.5	43	4.5	43	60	16	1:1	2:1	"
037	500 ct	500 ct	---	1.0	75	11	4.5	43	4 5	43	60	---	1.1	---	B 2/
038	500 ct	600 ct	150	1.0	.75	.11	4.5	43	4 5	43	66	17	1.1.1	1.83.1	A
[illegible]	600 ct	12.8 ct	3.2	1.0	.75	.11	4.0	68	4.0	68	1.75	44	6.89.1	13.7·1	"
	600 ct	30 ct	7 5	1 0	75	11	4 0	60	4 0	60	3 7	.93	4 47:1	8 94.1	"
[illegible]	600 ct	48 ct	12	1.0	.75	.11	4 0	60	4.0	60	5.9	1.5	3 54:1	7.07:1	"
042	600 ct	200 ct	50	1.0	.75	.11	4.0	60	4 0	60	25	6.3	1 73·1	3 46:1	"
043	600 ct	600 ct	150	1.0	.75	.11	4.0	60	4 0	60	66	17	1.1	2.1	"
044	600 ct	600 ct	---	1.0	.75	.11	4.0	60	4.0	60	68	---	1:1	---	B 2/
045	600 ct	900 ct	---	1.0	.75	.11	4.0	60	4.0	60	104	---	1:1.22	---	B 2/
046	600 ct	1200 ct	300	1.0	.75	.11	4 0	60	4 0	60	116	29	1·1.41	1.41:1	A
047	640 ct	12.8 ct	3.2	1.0	75	.11	4.0	69	4.0	69	1.75	44	7 07:1	14.14:1	"
048	700 ct	100 ct	25	1 0	75	11	4.0	62	4 0	62	10 6	2.7	2.65.1	5.29.1	"
049	800 ct	12.8 ct	3.2	1.0	.75	.11	3.5	78	3 5	78	1.75	44	7.91.1	15.8:1	"
050	800 ct	48 ct	12	1.0	.75	11	3.5	74	3.5	74	6	1.5	4.08 1	8.16.1	"

see footnotes at end of table

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TABLE I Electrical ratings - Continued 1/

Dash no	Primary impedance (ohms)	Secondary impedance (ohms)		Power watts			Primary unbalanced dc current (mA)	Primary dc resistance (ohms)	Secondary dc resistance (ohms)		Turns ratio primary/secondary		Circuit diagram configuration	
		Series	Parallel	1 kHz and higher	400 Hz and higher	150 Hz and higher			Series connection	Parallel connection	Secondary series connection	Secondary parallel connection		
051	800 ct	800 ct	200	1.0	.75	.11	3.5	74	102	26	1:1	2:1	A	
052	900 ct	600 ct	150	1.0	.75	.11	3.0	90	66	17	1:1.23:1	2:45:1	•	
053	900 ct	900 ct	---	1.0	.75	.11	3.0	90	102	--	1:1	---	B	2/
054	1000 ct	12 ct	3	1.0	.75	.11	3.0	115	1.75	44	9.13:1	18.26:1	A	
055	1000 ct	50 ct	12.5	1.0	.75	.11	3.0	100	5.8	1.5	4.47:1	8.94:1	•	
056	1000 ct	1000 ct	250	1.0	.75	.11	3.0	75	103	26	1:1	2:1	•	
057	1060 ct	12.8 ct	3.2	1.0	.75	.11	3.0	115	1.75	.44	9.1:1	18.2:1	•	
058	1200 ct	12.8 ct	3.2	1.0	.75	.11	3.0	122	1.75	.44	9.68:1	19.4:1	•	
059	1200 ct	500 ct	125	1.0	.75	.11	3.0	113	63	16	1.55:1	3.1:1	•	
060	1200 ct	1200 ct	300	1.0	.75	.11	3.0	113	155	39	1:1	2:1	•	
061	1200 ct	1200 ct	---	1.0	.75	.11	3.0	109	190	---	1:1	---	B	2/
062	1500 ct	12 ct	3	1.0	.75	.11	2.5	140	1.75	44	11.18:1	22.36:1	A	
063	1500 ct	500 ct	125	1.0	.75	.11	2.5	150	60	15	1.73:1	3.46:1	A	
064	1500 ct	600 ct	---	1.0	.75	.11	2.5	150	66	---	1.58:1	---	B	2/
065	1500 ct	1500 ct	---	1.0	.75	.11	2.5	150	168	---	1:1	---	B	2/
066	1800 ct	12.8 ct	3.2	1.0	.75	.11	2.5	140	1.75	.44	11.18:1	22.36:1	A	
067	2000 ct	8 ct	2	1.0	.75	.11	2.0	250	1.0	.25	15.8:1	31.6:1	•	
068	2000 ct	2000 ct	500	1.0	.75	.11	2.0	185	255	64	1:1	2:1	•	
069	2000 ct	2000 ct	---	1.0	.75	.11	2.0	175	240	--	1:1	---	B	2/
070	2000 ct	8000 ct	2000	1.0	.75	.11	2.0	185	1015	254	1.2	1.1	A	
071	2500 ct	2500 ct	625	1.0	.75	.11	2.0	245	268	68	1:1	2:1	•	
072	3000 ct	3000 ct	750	1.0	.75	.11	1.8	267	295	74	1:1	2:1	•	
073	3000 ct	3000 ct	---	1.0	.75	.11	1.8	267	368	--	1:1	---	B	2/
074	3800 ct	3800 ct	900	1.0	.75	.11	1.5	300	413	103	1:1	2:1	A	
075	4000 ct	16 ct	4	1.0	.75	.11	1.5	395	2.5	65	15.8:1	31.6:1	•	
076	4000 ct	800 ct	150	1.0	.75	.11	1.5	395	66	17	2.58:1	5.16:1	•	
077	4000 ct	1000 ct	250	1.0	.75	.11	1.5	395	108	27	2:1	4:1	•	
078	4000 ct	8000 ct	---	1.0	.75	.11	1.5	390	955	--	1:1.41	---	B	2/
079	5000 ct	500 ct	125	1.0	.75	.11	1.3	450	62	16	3.16:1	6.32:1	A	
080	5000 ct	5000 ct	1250	1.0	.75	.11	1.3	450	620	155	1:1	2:1	•	
081	7500 ct	50 ct	12.5	1.0	.75	.11	1.0	670	6	1.5	12.25:1	24.5:1	•	
082	7500 ct	100 ct	25	1.0	.75	.11	1.0	670	11	2.8	8.66:1	17.32:1	•	
083	7500 ct	600 ct	150	1.0	.75	.11	1.0	670	66	16.5	3.54:1	7.07:1	•	
084	7500 ct	2500 ct	625	1.0	.75	.11	1.0	670	268	67	1.73:1	3.46:1	•	
085	7500 ct	7500 ct	---	1.0	.75	.11	1.0	670	925	--	1:1	---	B	2/
086	8000 ct	12.8 ct	3.2	1.0	.75	.11	1.0	765	1.75	44	25:1	50:1	A	
087	8000 ct	1000 ct	250	1.0	.75	.11	1.0	695	106	26.5	2.83:1	5.66:1	A	
088	8000 ct	1200 ct	---	1.0	.75	.11	1.0	715	153	---	2.58:1	---	B	2/
089	9000 ct	9000 ct	---	1.0	.75	.11	1.0	925	1020	---	1:1	---	B	2/
090	10,000 ct	12.8 ct	3.2	1.0	.75	.11	1.0	768	1.6	.4	28:1	56:1	A	
091	10,000 ct	16 ct	4	1.0	.75	.11	1.0	756	1.7	43	25:1	50:1	•	
092	10,000 ct	100 ct	25	1.0	.75	11	1.0	780	10	2.5	10:1	20:1	•	
093	10,000 ct	200 ct	50	1.0	.75	.11	1.0	780	24	6	7.07:1	14.14:1	•	
094	10,000 ct	500 ct	125	1.0	.75	.11	1.0	780	60	15	4.47:1	8.94:1	•	
095	10,000 ct	600 ct	150	1.0	.75	.11	1.0	780	83	21	4.08:1	8.16:1	•	

See footnotes at end of table.

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TABLE I. Electrical ratings - Continued 1/

Dash no.	Primary impedance (ohms)	Secondary impedance (ohms)		Power watts			Primary unbalanced dc current (mA)	Primary dc resistance (ohms)	Secondary dc resistance (ohms)		Turns ratio primary/secondary		Circuit diagram configuration	
		Series	Parallel	1 kHz and higher	400 Hz and higher	150 Hz and higher			Series connection	Parallel connection	Secondary series connection	Secondary parallel connection		
096	10,000 ct	1000 ct	250	1.0	.75	.11	1.0	780	106	27	3.16:1	6.32:1	A	
097	10,000 ct	1200 ct	300	1.0	.75	.11	1.0	780	145	36	2.89.1	5.77:1	"	
098	10,000 ct	1500 ct	375	1.0	.75	.11	1.0	780	160	40	2.58:1	5.16:1	"	
099	10,000 ct	2000 ct	500	1.0	75	.11	1.0	780	300	75	2.24·1	4.47.1	"	
100	10,000 ct	5000 ct	1250	1.0	.75	.11	1.0	780	610	153	1 41.1	2.83:1	"	
101	10,000 ct	7500 ct	1875	1.0	75	11	1.0	780	932	233	1.16.1	2.31:1	"	
102	10,000 ct	10,000 ct	2500	1.0	.75	.11	1.0	780	1075	269	1.1	2:1	"	
103	10,000 ct	10,000 ct	---	1.0	.75	.11	1.0	715	985	---	1:1	---	B 2/	(8)
104	15,000 ct	400 ct	100	.66	66	.11	0.8	1165	52	13	6.12:1	12.25:1	A	
105	15,000 ct	500 ct	125	.66	.66	.11	0.8	1165	60	15	5.48:1	10.95:1	"	
106	15,000 ct	600 ct	150	.66	.66	.11	0.8	1165	80	20	5:1	10:1	"	
107	15,000 ct	10,000 ct	2500	.66	.66	.11	0.8	1165	1050	263	1.22:1	2.45:1	A	
108	15,000 ct	15,000 ct	3750	.66	.66	.11	0.8	1165	1600	400	1:1	2:1	"	
109	16,000 ct	1000 ct	250	.62	.62	.11	0.8	1560	108	27	4·1	8·1	"	
110	20,000 ct	500 ct	125	.5	.5	.11	0.7	1780	60	13	6.32:1	12.65:1	"	
111	20,000 ct	800 ct	150	.5	.5	.11	0.7	1750	82	20.5	5.77:1	11 55:1	"	
112	20,000 ct	800 ct	200	.5	.5	.11	0.7	1750	98	24.5	5:1	10·1	"	
113	20,000 ct	1000 ct	250	.5	.5	.11	0.7	1750	135	34	4.47:1	8.94:1	"	
114	20,000 ct	5000 ct	1250	5	.5	.11	0.7	1750	610	153	2:1	4.1	"	
115	20,000 ct	10,000 ct	2500	.5	.5	.11	0.7	1750	1075	269	1.41:1	2.83:1	"	
116	25,000 ct	500 ct	125	.4	.4	.11	0.6	2470	60	15	7.07:1	14.14:1	"	
117	25,000 ct	1000 ct	250	.4	.4	.11	0.6	2470	106	27	5:1	10:1	"	
118	30,000 ct	50 ct	12.5	.33	.33	.11	0.5	2730	6.2	1.6	24.5:1	49 :1	"	
119	30,000 ct	1200 ct	300	.33	.33	.11	0 5	2815	156	39	5:1	10:1	"	
120	30,000 ct	500 ct	125	.33	.33	.11	0.5	2730	61	15.3	7.7:1	15.5:1	"	
121	30,000 ct	200 ct	50	.33	.33	.11	0.5	2730	24	6	12.2:1	24.5:1	"	
122	30,000 ct	5000 ct	1250	.33	.33	.11	0.5	2730	620	155	2.45.1	4.9:1	"	
123	40,000 ct	400 ct	100	.25	.25	.11	0.5	3150	53	13	10:1	20:1	"	(8)

1/ Qualification testing and approval to M27/358-043 and M27/358-123 shall be sufficient to grant qualification approval to M27/358-001 through M27/358-123.

2/ Includes electrostatic shield. Shield terminal shall be number 4. Voltage ratio shall be 2 to 1 at 20 kHz.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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